Lambda KV-Cache Coprocessor

Micro-Architecture & Programming Specification

Longhorn Silicon (Team A71) — SSCS Chipathon 2026, Track D
`lambda_kv_coproc` · GF180MCU core-only macro

Revision A — generated from RTL at signoff

This document is the practical reference for *talking to* the hardened `lambda_kv_coproc` block over its four-wire SPI link: pinout, clocking, the byte-level protocol, the address map and data formats, the status/decision bit-fields, the compress-step state machine, and a complete worked transaction. All values are taken directly from the RTL as hardened (`src/lambda_kv_coproc.sv`, `src/spi_loader.sv`); the hardened tile is D=2, L=2.

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1 Overview and interfaces

The coprocessor performs one *compress step* per invocation: the host streams L value tokens (D `fp16` channels each) and L per-token attention masses in over SPI; three engines then run and the host reads the results back. The engines are:

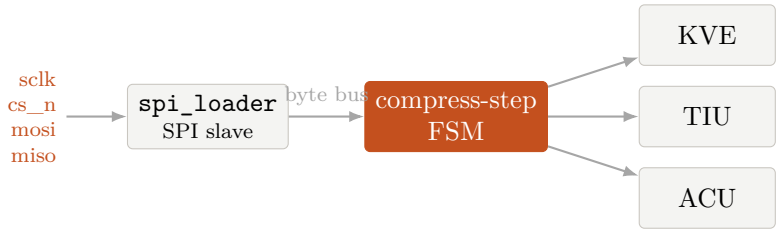
- **KVE** — Walsh–Hadamard rotation + per-token amax + INT3 quantization of each value token (codes, scale, reconstruction).
- **TIU** — H2O importance accumulation, a keep-tier bitmap, and a single eviction victim.
- **ACU** — a divide-free per-tile precision gate.

1.1 Top-level ports

Port	Dir	Function
`clk`	in	core clock (see §2)
`rst_n`	in	active-low synchronous-release reset
`spi_sclk`	in	SPI serial clock (host)
`spi_cs_n`	in	SPI frame select, active low
`spi_mosi`	in	host → chip data
`spi_miso`	out[1]	chip → host data
`obs0..obs3`	out[1]	status taps: busy, done, err, gate_fp16
`VDD/VSS`	pwr	5 V core power / common ground

Build parameters (hardened values): `D=2` (value channels/token, power of two), `L=2` (cache slots/-tokens per step), `ADDR_WIDTH=16`. Derived constants below assume these values; where a formula helps, L and D are kept symbolic.

1.2 Internal block map



The `spi_loader` decodes SPI frames into a byte-addressed bus (`bus_addr`, `bus_wdata`, `bus_we`, `bus_re`, `bus_rdata`) plus a `start` pulse; it feeds `status_in` back to the host. The compress-step FSM owns the buffers and sequences the engines. Interfaces:

Engine (module)	Inputs from FSM	Outputs to FSM
KVE `cq_value_path_wht_syn`	`in_vec`[$16D$], `dec_idx`	`out_codes`[$8D$], `out_scale`, `dec_rot_f16`
TIU `token_importance_unit`	`acc_*`, `ld_*`, `evict_req`, `tier_threshold`=48	`evict_valid/slot`, `tier_keep`, `busy`
ACU `precision_controller`	`s_valid/data/last`	`d_valid`, `d_fp16`

The KVE is combinational; TIU and ACU are clocked and handshaked by the FSM.

[1] `spi_miso` and `obs0..obs3` are physically bidirectional pads (no output-only digital pad exists in the GF180 IO library) used as outputs.

2 Clocking, reset, and timing budget

Core clock. The block is hardened at a **300 ns period (≈3.33 MHz)**; this bound is set by the combinational Walsh–Hadamard value path and is the maximum core-clock frequency. Timing closes with zero setup/hold violations at this period.

Reset. `rst_n` is active low with synchronous release. Hold it low for several core clocks after power-good; keep `spi_cs_n` high (idle) during reset. On release the FSM is in `S_IDLE` with all status bits clear.

SPI vs. core clock (important). The SPI inputs are *oversampled in the core clock domain* through a 3-flop synchronizer with rising-edge detection; there is no dedicated async SPI front end. Correct operation therefore requires

$$f(\texttt{spi\_sclk}) \ll f(\texttt{clk}), \qquad T_{\texttt{sclk-phase}} \gtrsim 4\,T_{\texttt{clk}}.$$

In practice budget ≥ 8 **core clocks per SPI bit**. With the core at its 3.33 MHz ceiling this caps `spi_sclk` at roughly **400 kHz**; run slower (100–250 kHz) for margin. `spi_cs_n` must stay low for the whole frame and be sampled by at least two core clocks at each edge.

At-a-glance status. Independently of SPI, the observation pins mirror the status flags live: `obs0`=busy, `obs1`=done, `obs2`=err, `obs3`=gate_fp16 — useful on a logic analyzer or LEDs during bring-up.

3 SPI protocol

Mode 0 (CPOL=0, CPHA=0), MSB-first. A frame is delimited by `spi_cs_n` low → high. Data changes while `spi_sclk` is low and is sampled by the slave on the *rising* edge; the host samples `spi_miso` in the high phase. MISO is driven combinationally and is valid for the whole byte, so the host may sample early in the high phase.

Frame layout.

Byte	Field	Notes
0	`CMD`	command (below)
1	`ADDR[15:8]`	big-endian address, high byte first
2	`ADDR[7:0]`	
3..n	`DATA`	payload; **address auto-increments per byte**

Command set.

Value	Command	Behaviour
`0x01`	`WRITE`	write `DATA` bytes starting at `ADDR` (auto-inc)
`0x02`	`READ`	return `DATA` bytes from `ADDR` on MISO (auto-inc)
`0x03`	`START`	pulse `start`; run one compress step (no `ADDR`/`DATA`)
`0x04`	`STATUS`	MISO returns the STATUS register directly

For `READ`, after `CMD`+2 address bytes the host clocks n dummy bytes (e.g. `0x00`) on MOSI and captures the returned bytes on MISO; the internal address advances at each byte boundary, so a single frame streams a whole region. `START` is a one-byte frame. Reading STATUS is normally done with `READ` @ `0x0001` (below); `CMD_STATUS` is an equivalent shortcut.

4 Address map and data formats

16-bit byte address space; multi-byte fields are **little-endian on the wire** (low byte first). Element ordering is token-major: token 0's D channels, then token 1, etc. `fp16` = IEEE half precision.

Address	Region	Size (bytes)	Dir
`0x0300`	`W` — per-token attention mass (u8)	L	write
`0x0400`	`V` — value tokens (fp16)	$2LD$	write
`0x0800`	`CODES` — rotated INT3 codes (int8)	LD	read
`0x0A00`	`SCALE` — per-token scale (fp16)	$2L$	read
`0x0C00`	`VHAT` — rotated reconstruction (fp16)	$2LD$	read
`0x0001`	`STATUS`	1	read
`0x0002`	`DECISION`	1	read
`0x0003`	`KEEP` — keep-tier bitmap	1	read

For the hardened tile (D=2, L=2): `W`=2 B, `V`=8 B, `CODES`=4 B, `SCALE`=4 B, `VHAT`=8 B. `V` byte order at `0x0400`: token0-ch0 low, token0-ch0 high, token0-ch1 low, token0-ch1 high, token1-ch0 low, …

4.1 Status / decision / keep bit-fields

`STATUS` (`0x0001`, read-only):

bit 7	6	5	4	3	2	1	0
0	`err`	`gate_fp16`	0	0	0	`done`	`busy`

`DECISION` (`0x0002`, read-only):

bit 7	bits $[\lceil \log_2 L \rceil - 1 : 0]$
`gate_fp16`	`evict_slot`

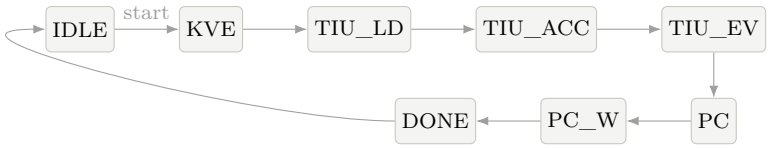
For L=2: bit 7 = `gate_fp16`, bit 0 = `evict_slot` (bits 6..1 = 0).

`KEEP` (`0x0003`, read-only): low L bits are the keep-tier bitmap; bit s is set when slot s's accumulated mass $\geq$ `tier_threshold` (=48). Upper bits read 0.

5 Compress-step state machine

On `START` the FSM leaves `S_IDLE` (asserting `busy`) and walks the sequence below, ending in `S_DONE` (assert `done`, deassert `busy`) and returning to idle. Encodings are the RTL's 4-bit state values.

Enc	State	Action
0	S_IDLE	wait for start; clear done/err.
1	S_KVE	for each token, sweep D channels: latch INT3 code and fp16 reconstruction; capture per-token scale at channel 0.
2	S_TIU_LD	install L slots into the TIU.
3	S_TIU_ACC	accumulate each slot's attention mass.
4	S_TIU_EV	request eviction victim; latch keep-tier bitmap.
5	S_PC	stream the masses into the ACU as the score tile.
6	S_PC_W	wait for the ACU decision; latch gate_fp16.
7	S_DONE	assert done, deassert busy; → idle.



Error / timeout. S_TIU_EV bounds the eviction handshake and proceeds even if it times out. S_PC_W bounds the ACU handshake; if the decision does not arrive within its window it sets err (STATUS bit 6) and goes to S_DONE. A run with err=1 means the ACU did not respond; done still asserts so the host's poll terminates.

6 Address decode

All decode is combinational in the coprocessor:

- **Write:** bus_we writes W when bus_addr$\in$ [0x0300, 0x0300+L) and V when $\in$ [0x0400, 0x0400+2LD); for V the low address bit selects the low/high fp16 byte.
- **Read mux:** 0x0001→STATUS, 0x0002→DECISION, 0x0003→KEEP, CODES/SCALE/VHAT from their windows; any other address reads 0x00.

7 Programming sequence (worked example)

One full compress step for D=2, L=2. Each line is one SPI frame (CS low … high). Bytes are hex; -- marks captured MISO bytes.

```
; 1. write attention masses W (2 bytes) at 0x0300
WRITE 0x0300  m0 m1                       ; CMD=01 ADRH=03 ADRL=00 DATA=m0,m1

; 2. write value tokens V (fp16 LE, 8 bytes) at 0x0400
WRITE 0x0400  t0c0L t0c0H t0c1L t0c1H
              t1c0L t1c0H t1c1L t1c1H

; 3. kick one compress step
START                                     ; CMD=03 (no address/data)
```

```
; 4. poll STATUS until done (bit1), bounded; check err (bit6)==0
loop: READ 0x0001 -> s                    ; CMD=02 ADRH=00 ADRL=01, 1 dummy byte
      until (s & 0x02)                     ; done
      assert (s & 0x40) == 0               ; err clear

; 5. read the decision + keep bitmap
READ 0x0002 -> d      ; bit7=gate_fp16, bit0=evict_slot (L=2)
READ 0x0003 -> keep   ; bits[1:0] = keep-tier per slot

; 6. read the compressed records
READ 0x0800 -> c0 c1 c2 c3                ; CODES  (L*D = 4 int8, range [-4,3])
READ 0x0A00 -> s0L s0H s1L s1H            ; SCALE  (L fp16)
READ 0x0C00 -> h0L h0H ... h3L h3H        ; VHAT   (L*D fp16)
```

A `READ` frame sends `CMD=02`, the two address bytes, then n dummy MOSI bytes while capturing n MISO bytes; the address auto-increments, so the whole region streams in one frame.

8 Caveats and known issues

- **Reserved low addresses `0x0000`–`0x0005`.** The loader keeps vestigial local CSR shadows here from an earlier datapath. A `WRITE` to `0x0000` pulses `start` (bit 0) and latches `precision_sel` (bits [2:1]); writes to `0x0002`–`0x0005` land in unused local registers. **Do not write `0x0000`–`0x0005`**; use `CMD_START` to start. (Reads at `0x0001`/`0x0002`/`0x0003` return the coprocessor STATUS/DECISION/ KEEP as documented.)
- **Loader front end.** The SPI slave oversamples in the core domain and is documented in-RTL as a pre-tapeout skeleton (no metastability-hardened async front end). Respect the `spi_sclk` $\ll$ `clk` budget of §2.
- **Scope.** Verified at the RTL + physical-signoff level (self-checking SPI test + LVS). No gate-level sim; the SPI test is a targeted deterministic check, not exhaustive coverage. See `docs/VERIFICATION.md`.

Source of truth: `src/lambda_kv_coproc.sv`, `src/spi_loader.sv` on `main`. Functional model: `src/tb_coproc_spi.sv` (`make sim-coproc`).